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Roll No

ME-705-GS

B.E. VII Semester

Examination, December 2020

Grading System (GS)

Operations Research and Supply Chain

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Explain degeneracy in transportation problem.

b) Find the minimum value of

$$Z = -x_1 + 2x_2$$

Subjected to: $-x_1 + 3x_2 \leq 10$

$$x_1 + x_2 \leq 6$$

$$x_1 - x_2 \leq 2$$

$$x_1, x_2 \geq 0$$

2. a) What is Supply Chain Management? Discuss its importance.

b) Explain the role of information technology in supply chain management.

3. What is Inventory? What is necessity to Maintain Inventory? Discuss the concept of economic order quantity.

4. What is ABC and VED system of inventory analysis?

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5. The following table gives the annual demand and unit page of 4 items.

Item	A	B	C	D
Annual demand (units)	800	400	392	13,800
Unit page (Rs)	0.02	1.0	8.00	0.20

Order cost is Rs 5 per order and holding cost is 10% of unit cost. Determine

- i) EOQ of each item
 - ii) Total variable cost
 - iii) EOQ in Rs
 - iv) EOQ in years of supply and No. of orders of each item separately.
6. a) Explain AHP and discuss the procedure of AHP model.
b) What is vendor rating? Give the various criterion on which vendor rating depends.
7. a) Discuss the impact of Bull-whip effect and outline how it leads to increased risk and poor service level. Comments on remedies to mitigate the impact of this phenomenon.
b) What are the essential characteristics of a linear programming model? Also define slack and surplus variables in LPP.
8. Write a short note on the following.
a) Heuristic and meta heuristic algorithm
b) Assignment Problems
